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EEG Singal Classification in Neurological Disorders

P R E S E N T A T I O N 2 0 2 6

 Student Name
R Gnana Deepika

 Student Email
ramalagnanadeepika@gmail.com

 Student Name
N Bhavana

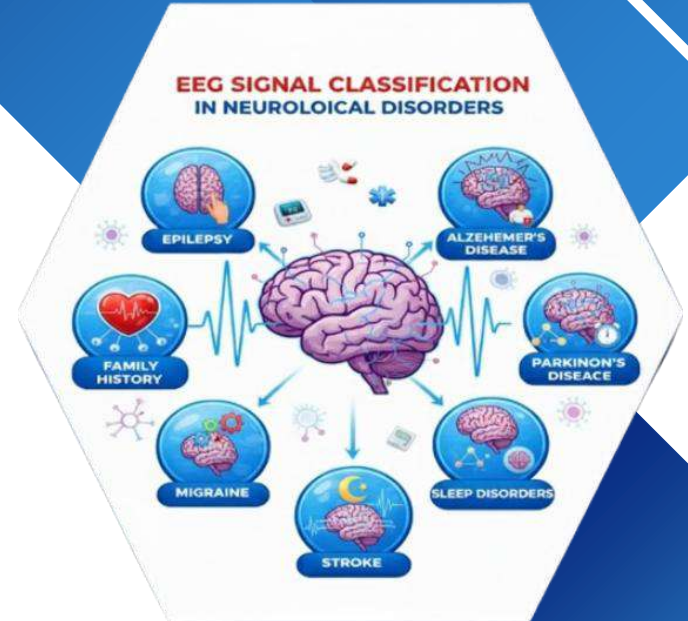
 Student Email
bhavanaists78@gmail.com

 Student Name
V Umadevi

 Student Email
umadeviists23@gmail.com

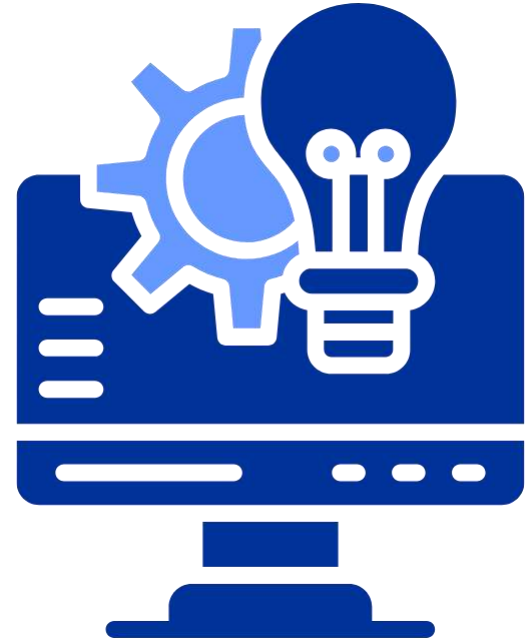
 Student Name
P Tejaswini

 Student Email
palisaitejaswini@gmail.com



OUTLINE

- Abstract of the Project
- Problem Statement
- Proposed Solution
- System Architecture
- Live Demo of the Project
- Embedded Video of Project
- Conclusion
- Future Scope



- **Abstract:**

EEG signals are used to analyze brain activity and diagnose neurological and mental health disorders. Manual interpretation of EEG data is complex, time consuming and prone to human error.

The proposed system uses a Deep Learning- based LSTM model for automatic multi- class classification. It detects multiple disorders including Anxiety, OCD, Addictive disorders and healthy conditions. EEG signals are preprocessed using filtering , normalization and segmentation to improve accuracy. An LLM based explanation module provides human readable interpretations of predictions.



Problem Statement :

EEG data is highly complex and difficult to interpret manually .

Manual diagnosis depends heavily on expert neurologists.

Large volumes of EEG data increase analysis time and effort .

Human errors may occur during manual interpretation.

Multiple neurological disorders show overlapping EEG patterns.

Existing automated systems lack explainability and transparency



Proposed Solution

Develop an automated EEG classification system using a Deep Learning –based LSTM model. Preprocess raw EEG signals through filtering, normalization and segmentation to improve data quality Train the LSTM model to perform accurate multi-class classification of neurological disorders. Integrate an LLM-based module to generate clear and human –readable explanations for predictions. Implement a private blockchain using Ganache to ensure and tamper- proof result storage. Design a web-based client –server application for real-time EEG data upload prediction visualization



System Architecture

User Interface Layer

A web-based interface built using HTML and CSS allows users to upload EEG data, view predictions, and access explanations in real time.

AI & Processing Layer:

EEG signals are preprocessed and fed into the LSTM model for multi-class classification, while an LLM module generates prediction explanations.

Backend Application Layer

Developed using Flask, this layer handles file uploads, preprocessing, model execution, explanation generation, and communication with the blockchain.

Blockchain & Storage Layer:

Classification results are securely stored using a private Ganache blockchain to ensure data integrity and tamper-proof record keeping.

Live Demo of Project

Here's the list of some demo of the projects -



Code

I recently explored a **heart disease risk prediction dataset** to understand the key factors affecting cardiovascular health. The goal was to identify patterns and insights that can help in **early detection and prevention**.



Input

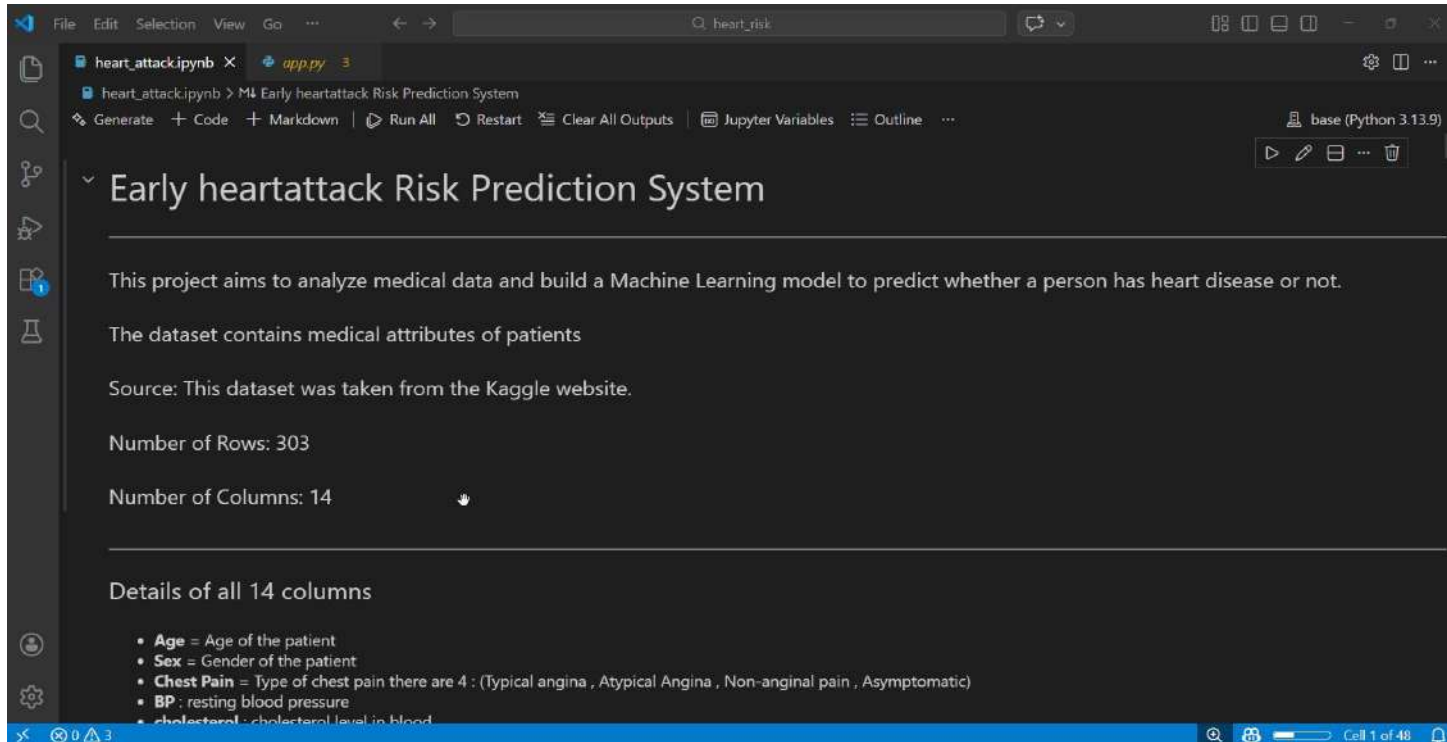
The inputs we have chosen for this dataset are Age, cholesterol , BP , chest pain.....



Output

The output is to predict the risk percentage of the heart attack based on the inputs and visualization

Video of Project Demo



The screenshot displays a Jupyter Notebook environment with a dark theme. The notebook is titled 'heart_attack.ipynb' and is part of a project named 'M4 Early heartattack Risk Prediction System'. The interface includes a top menu bar with options like File, Edit, Selection, View, and Go. Below the menu, there are tabs for 'heart_attack.ipynb' and 'app.py'. The notebook content is organized into sections:

- Early heartattack Risk Prediction System**
 - This project aims to analyze medical data and build a Machine Learning model to predict whether a person has heart disease or not.
 - The dataset contains medical attributes of patients
 - Source: This dataset was taken from the Kaggle website.
 - Number of Rows: 303
 - Number of Columns: 14
- Details of all 14 columns**
 - **Age** = Age of the patient
 - **Sex** = Gender of the patient
 - **Chest Pain** = Type of chest pain there are 4 : (Typical angina , Atypical Angina , Non-anginal pain , Asymptomatic)
 - **BP** : resting blood pressure
 - **cholesterol** : cholesterol level in blood

The bottom status bar indicates 'Cell 1 of 48'.

Conclusion

The proposed EEG signal classification system successfully automates the detection of multiple neurological and mental health disorders using an LSTM-based deep learning model. The system reduces manual effort, improves diagnostic accuracy, and efficiently handles complex EEG time-series data. By integrating an LLM-based explanation module, it enhances transparency and clinical trust in predictions. The use of a private blockchain ensures secure, tamper-proof storage of classification results, maintaining data integrity. Overall, the project combines deep learning, explainable AI, and blockchain technology to provide a reliable, secure, and intelligent framework for automated neurological disorder detection.



Future Scope

The proposed EEG classification system can be enhanced to support real-time monitoring and live disorder detection. The model can be expanded to identify additional neurological and psychiatric conditions with improved accuracy. Integration with hospital systems and cloud platforms can increase scalability and accessibility. Future improvements may also include advanced AI-based predictive analysis, mobile application support, and IoT-enabled EEG device integration for remote healthcare use.



Reference:

<https://www.kaggle.com/datasets/neurocipher/heartdisease> -

Kaggle

<https://github.com/Deepika-Ramala/Student-Projects/upload>

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Git Hub



Thank You